SIT305 Task 4.0 HD
Android App Proposal

Bowerbird: Automatic Image Organiser

Introduction

Bowerbird is an automatic image organisation tool, harnessing on-device AI to privately analyse images with user-defined prompts. By applying bespoke rules to watched folders, users can personalise image management to match their own thought processes.

Primary competitors

Google Photos - Simple AI tagging and internal organisation without customisation.

Tasker – Complex automated rule system for a wide range of applications, excluding image analysis.

Gallery Sort, Pico, etc. – Dedicated photo organisers, using manual swipe-to-organise.

Objective

Smartphone users generally organise images by assigning them to albums manually, or using services such as Google Photos to automatically create collections of similar photos (e.g. photos of a specific person). This forces users to make a choice between convenience and specificity. Bowerbird aims to achieve both, providing *automated specificity*.

Proposed Solution

Bowerbird organises images using **rules** applied to watched folders. A rule is a statement comprised of one or more conditions triggering one or more actions. For example, a rule "*Sort cats*" might be defined as "*If the image contains a cat, then move it to /Pictures/Cats*". If the "*Sort cats*" rule were applied to the */DCIM/Camera* directory, when the user takes a photo of a cat, it is moved to */Pictures/Cats*.

AI use

Gemini Nano's **multimodal prompts** [1] allow for an image and associated query to be passed as a single prompt. Bowerbird utilises this functionality both to analyse the image for conditions, and to perform actions. For example, "*If the image includes a dress and the vibe is vintage 1950's, then tag it with the style of the dress*".

Target audience

Users with many images needing to be organised. Bowerbird is designed to be user-friendly, but tech-savvy users will be more comfortable. With those traits in mind, the target audience is 25–40-year-old parents and creative professionals.

User stories

As a **parent**, I want to retrieve specific memories of my family, so that I can relive and share them with friends and family.

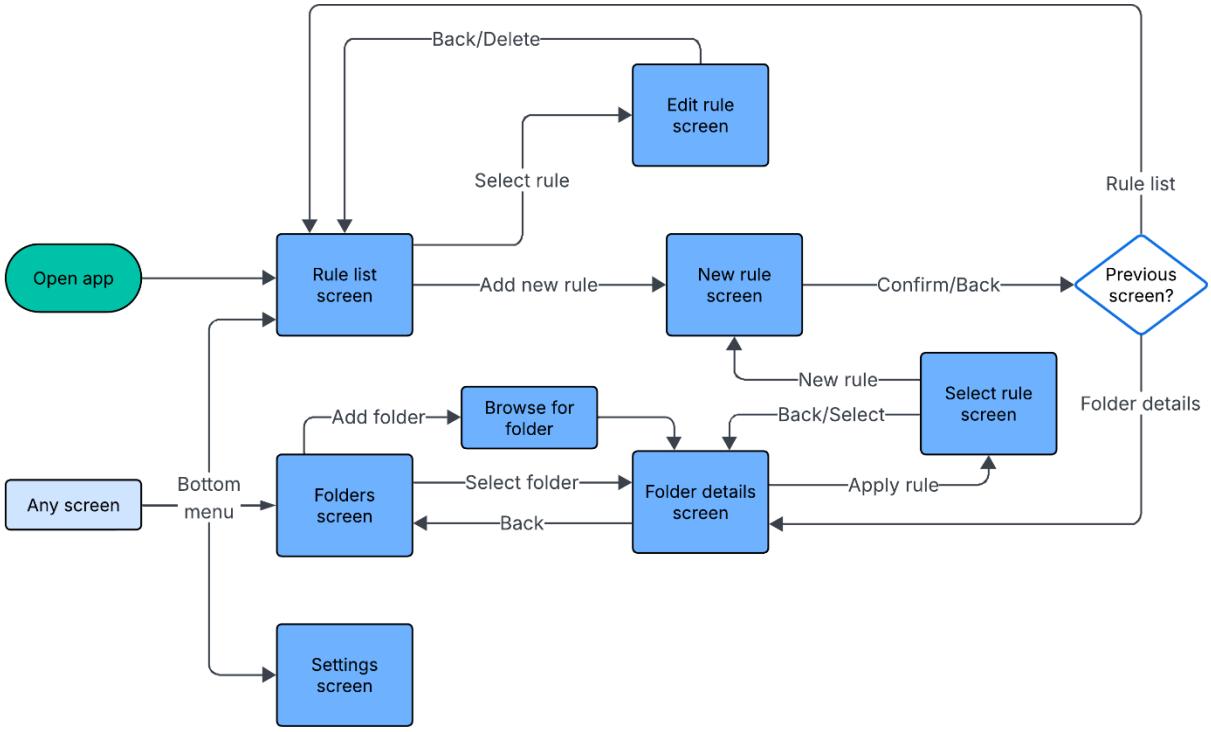
As a **creative professional**, I want to organise my research and inspiration images into groups, so that I can use them as a reference for my projects.

Core features

- AI-powered image analysis and tagging
- Image organisation (e.g. directory placement)
- Basic image adjustment (e.g. rotating)
- Text extraction
- Image analysis without privacy concerns

Screens

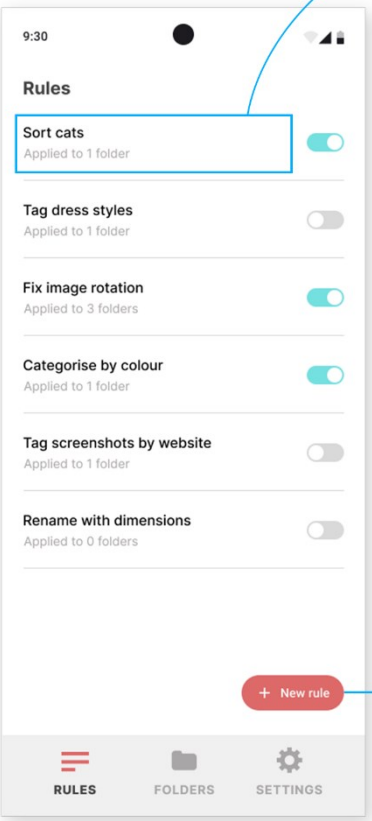
User flow diagram



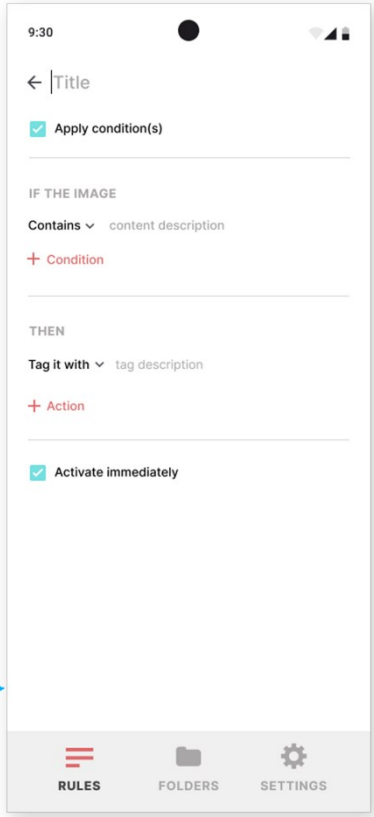
Shows the user's flow between screens as they navigate the app.

Example screen mock-ups

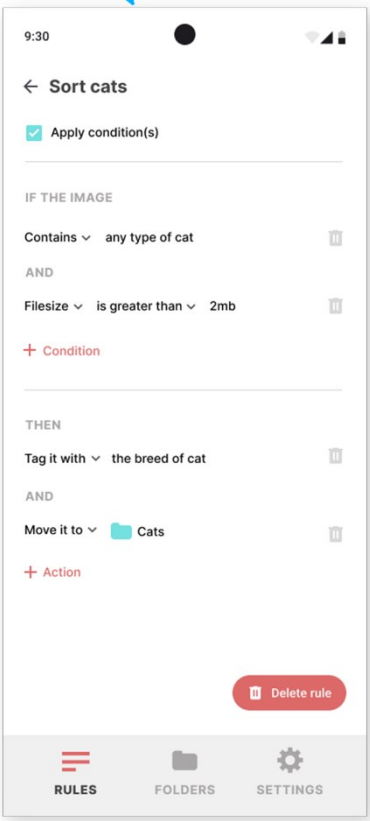
Rule list



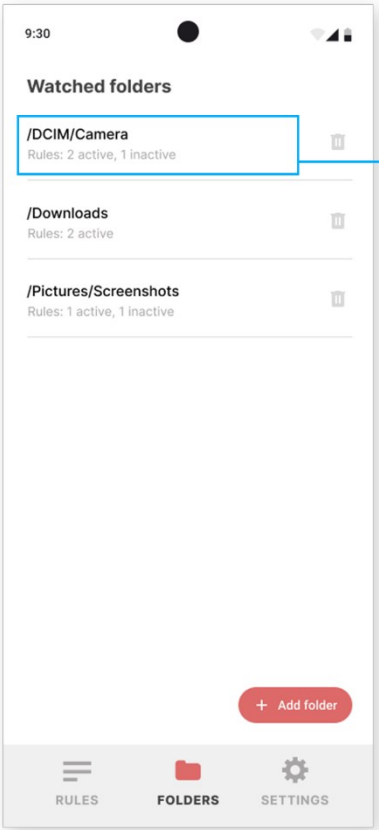
New rule



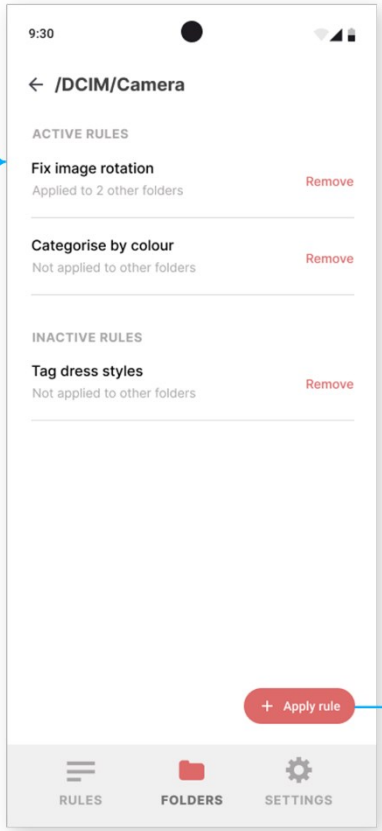
Edit rule



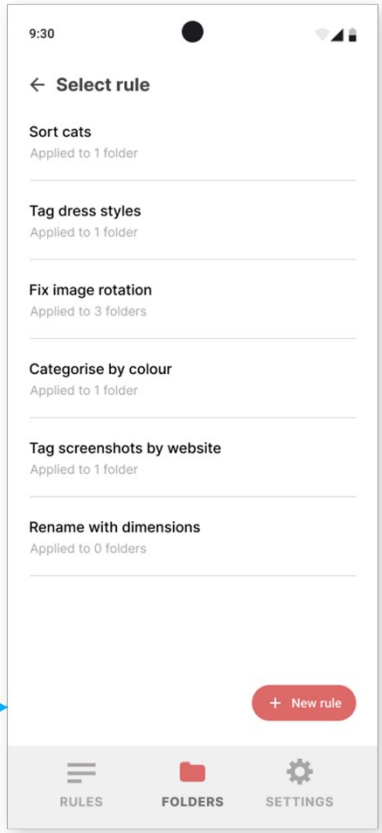
Folders



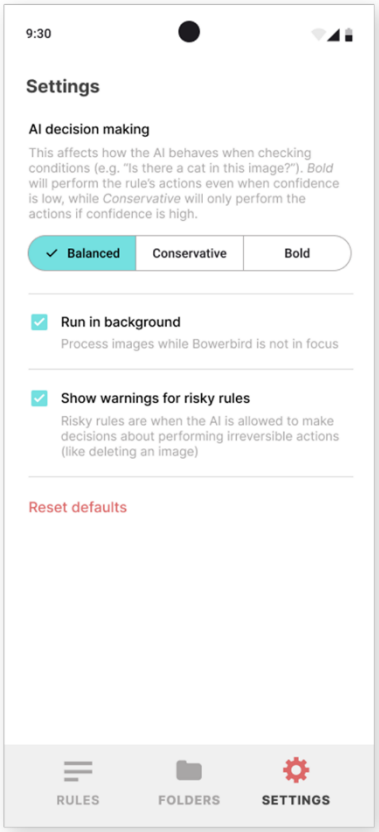
Folder details



Select rule



Settings



Example screens showing some common interactions.

Quality considerations

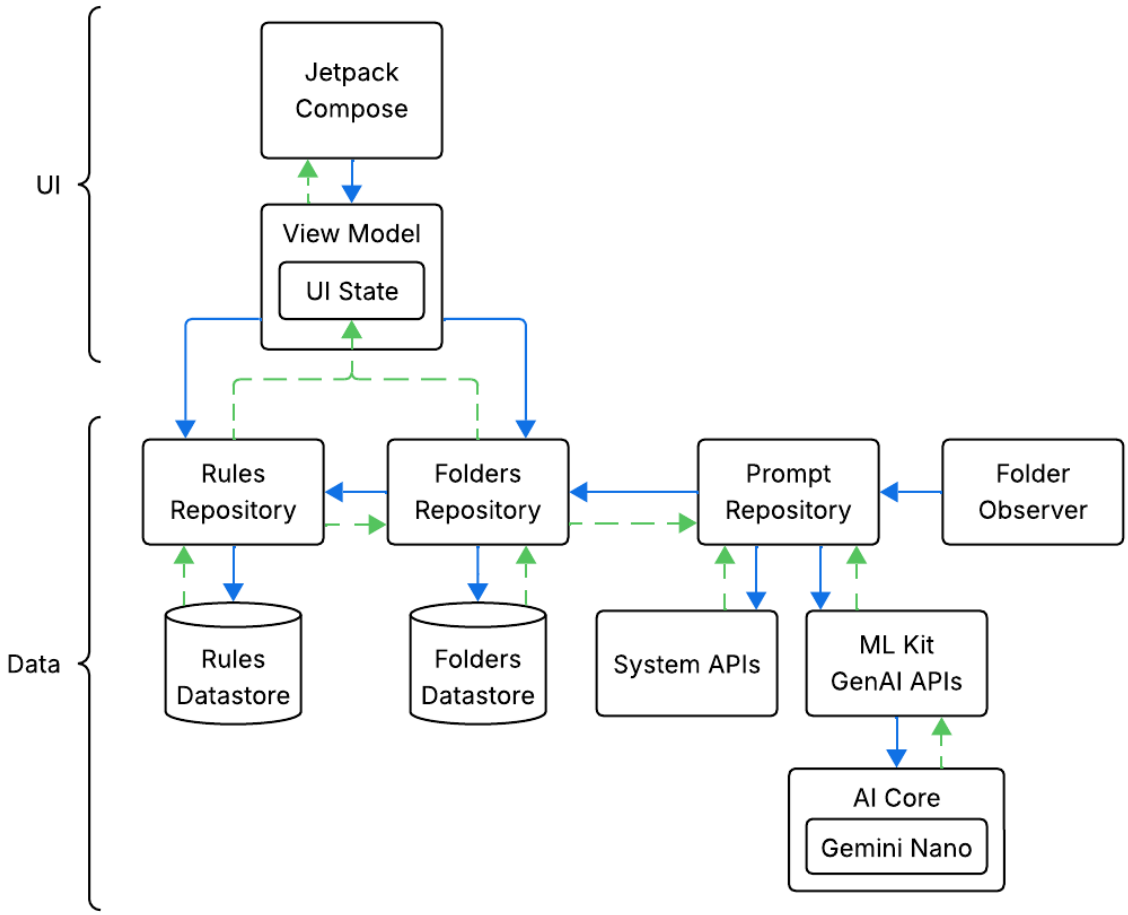
Safety controls are built into Gemini Nano itself [2]. Therefore, Bowerbird's role is to handle refusal cases appropriately. In this case, the rule will fail and the user will be notified.

Some actions may be irreversible (e.g. overwrite). When a "risky" rule is created (an irreversible action based on an AI-defined condition), the user is notified.

Technology Stack & Architecture Diagram

IDE: Android Studio
Languages: Kotlin
APIs: ML Kit GenAI APIs

Architecture diagram [3]



Prerequisites and assumptions

- On-device AI requires a modern, high-end device (Pixel 8 or newer), 8GB RAM minimum, recommended 12GB+ [4]
- Android 10+ installed [5]
- Explicit permissions required for some actions

Development Methodology

Bowerbird incorporates experimental AI features, and is broad enough to fill a range of needs. Therefore, an Agile approach will be taken, bringing flexibility where required based on stakeholder feedback. A test-driven approach is not mutually exclusive, and will be utilised for the core AI components to ensure responses are consistent and useful.

Conclusion

By utilising Gemini Nano's multimodal prompts, Bowerbird will allow users to automate image organisation with a bespoke approach not offered by other apps in the domain. With a broad range of applications and on-device processing for privacy, there is potential for a wide user base.

References

[1] Google Developers (2026). *Send custom requests to Gemini Nano with GenAI Prompt API* [Website]. Available: https://developers.google.com/ml-kit/genai/prompt/android

[2] Google Developers (2026). *Evaluate prompt quality* [Website]. Available: https://developers.google.com/ml-kit/genai/prompt/android/evaluate-prompt

[3] Android Developers (2026). *Guide to app architecture* [Website]. Available: https://developer.android.com/topic/architecture

[4] TechSpot (2024). *Google says running AI locally on phones requires tons of RAM* [Website]. Available: https://www.techspot.com/news/102450-google-running-gemini-ai-smartphones-require-gobs-ram.html

[5] Androidsis (2026). *What is Gemini Nano and what Android phones is it compatible with?* [Website]. Available: https://en.androidsis.com/What-is-Gemini-Nano-and-what-Android-phones-is-it-compatible-with%3F/